

Quantum Algorithm for Identifying Hidden Graphs

arXiv:2605.11228

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Before we start

- ▶ quantum computing needs **new algorithms**, and **new applications** for the ones we have
- ▶ the **Genesis Mission** is aimed at exactly that: using AI for quantum algorithm discovery
- ▶ $\Rightarrow$ the **paper** behind this talk, [arXiv:2605.11228](https://arxiv.org/abs/2605.11228), was a warm-up for it
 - ▶ the ideas are mine, found without AI
 - ▶ the numerics and the figures took non-trivial AI assistance

The problem

A Hamiltonian H_b depends on an unknown bit $b \in \{0, 1\}$.

- ▶ H_b **sparse** and **exponentially large**
- ▶ $|\psi\rangle$ a **computational basis state**, with signal

$$f_b(t) := \langle \psi | e^{-iH_b t} | \psi \rangle$$

- ▶ our **only access** to H_b is this signal
- ▶ *small* graphs A_0, A_1 and a vertex v , all known to us
- ▶ we are **promised** that, for all t ,

$$f_b(t) = \langle v | e^{-iA_b t} | v \rangle$$

Decide b .

Where the small graph comes from

- ▶ we will **build** H_b so that the states $H_b^k|\psi\rangle$ span only a small subspace — the **Krylov subspace** $\mathcal{K}$
- ▶ $\Rightarrow$ tiny inside the exponentially large space H_b acts on
- ▶ and so that on $\mathcal{K}$ the operator is not merely small — it *is* a graph:

$$\underbrace{H_b|_{\mathcal{K}}}_{\text{effective Krylov Hamiltonian}} = \underbrace{A_b}_{\text{adjacency matrix, small graph}}$$

- ▶ the starting state sits at a single vertex: $|\psi\rangle \leftrightarrow |v\rangle$

A simple quantum algorithm; classically hard

- ▶ both A_b are known $\Rightarrow$ compute both signals classically
- ▶ find the time at which they differ the most
- ▶ run the evolution at that time, and see which value comes back
- ▶ the small graphs say *when* to look and *which two values* to expect

Quantum side: easy

- ▶ controlled $e^{-iH_b t}$, with a Hadamard test to estimate the signal

Classical side: hard

- ▶ no way to evaluate a signal living in 2^N dimensions

The algorithm: the classical half

- ▶ **before the machine is on:** A_0 and A_1 are small, so this is ordinary linear algebra

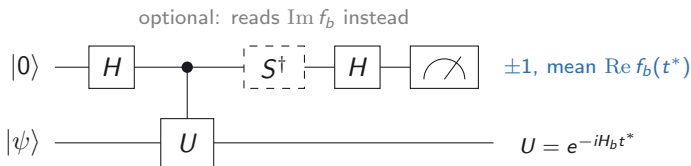
t^* = a time where the signals differ, $\text{Dis} = |f_0(t^*) - f_1(t^*)|$

- ▶ exact — no sampling, no approximation

Choosing t^* costs no quantum computer at all.

The algorithm: the quantum half

- ▶ **at $t = t^*$:** a **Hadamard test**, one controlled $e^{-iH_b t^*}$ per repetition
- ▶ $N_{\text{rep}} = \mathcal{O}(1/\text{Dis}^2)$ of them



Reading the signal needs the interferometer.

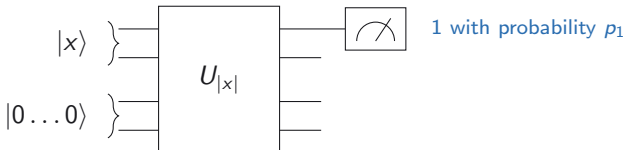
Two instances

	big Hamiltonian H_b	small graph A_b
BQP	clock $\otimes$ circuit	C_m vs. C_{2m}
spired graphs	spired prism / Möbius G_{spire}	towered graph G_{tower}

- ▶ **BQP** — deciding whether C_m or C_{2m} hides inside the Hamiltonian captures the *full power of quantum computation*.
A **theorem**.
- ▶ **spired graphs** — deciding whether a *prism* or a *Möbius ladder* hides inside the big obfuscated graph is **conjectured** to be classically hard, in the **black-box model**.

Formalizing the power of quantum computing

A language $L \subseteq \{0, 1\}^*$: given an input x , decide whether $x \in L$.



- ▶ the circuit $U_{|x|}$ is built from $\text{poly}(|x|)$ elementary gates, on N qubits
- ▶ one **output qubit** is measured $\Rightarrow$ outcome 1 with probability p_1
- ▶ **promise:** $x \in L \Rightarrow p_1 > 2/3$, $x \notin L \Rightarrow p_1 < 1/3$

That is BQP. One bit: accept or reject.

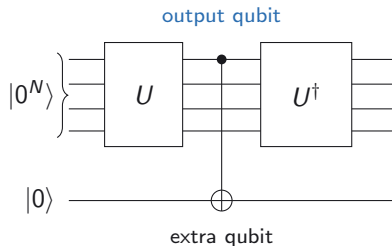
Two simplifications, both for free

- ▶ **Absorb the input.** Flip the qubits where x has a 1
 $\Rightarrow$ x is part of the circuit; the start is always $|0^N\rangle$
- ▶ **Amplify the gap.** Repeat and take the majority
 $\Rightarrow$ w.l.o.g. $p_1 = 0$ (**NO**) or $p_1 = 1$ (**YES**)

One circuit U , start $|0^N\rangle$, answer one bit b .

Step 1: from a circuit to an involution

- ▶ adjoin **one extra qubit**; the circuit copies its answer onto it, then undoes itself
- ▶ start from $|\text{init}\rangle = |0^N\rangle|0\rangle$



$$V = (U^\dagger \otimes I) \text{CNOT} (U \otimes I)$$

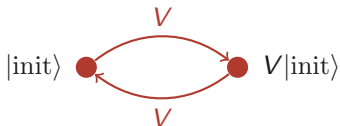
A Hermitian involution: $V = V^\dagger, \quad V^2 = I$

The answer is already a dimension

- ▶ $V^2 = I \Rightarrow$ the orbit of $|\text{init}\rangle$ is **one point or two**



NO ($p_1 = 0$): $\dim = 1$



YES ($p_1 = 1$): $\dim = 2$

- ▶ in the YES case the extra qubit flips, so V **swaps** two states instead of fixing one

One bit of dimension — and that bit is the circuit's answer.

Step 2: from an involution to a Hamiltonian

- ▶ factor $V = V_m \cdots V_1$ into m elementary gates, add a clock register of dimension m , and *walk*:

$$\underbrace{F}_{\text{forward-time}} = \sum_{t=0}^{m-1} |t+1 \bmod m\rangle\langle t|_c \otimes V_{t+1} \qquad H = F + F^\dagger$$

- ▶ H is $O(1)$ -**local**, with $\text{Spec}(H) \subseteq [-2, 2]$
- ▶ start from $|\text{init}\rangle = |0\rangle_c \otimes |0^N\rangle|0\rangle$

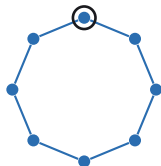
H has dimension $m \cdot 2^{N+1}$ — exponentially large.

The two cycles

One lap of the clock applies the whole of V :

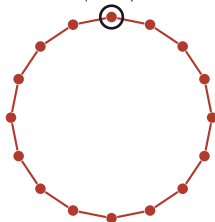
$$F^m|\text{init}\rangle = |0\rangle_c \otimes V|\text{init}\rangle$$

$|\text{init}\rangle$



NO: C_m
one lap comes back

$|\text{init}\rangle$



YES: C_{2m}
two laps to come back

- ▶ same local picture everywhere: every vertex has two neighbours
- ▶ $\Rightarrow$ the two cases differ only in **how far around we must go**
- ▶ $C_{2m} \rightarrow C_m$ is the $\mathbb{Z}_2$ **double cover**: the extra qubit is the second sheet

What only the quantum walk sees

- ▶ H_b acts on $m \cdot 2^{N+1}$ dimensions — far too big to write down
- ▶ but from $|\text{init}\rangle$ the evolution never leaves the orbit of F : m states in the NO case, $2m$ in the YES case
- ▶ the orbit is orthonormal and H_b -invariant $\Rightarrow$ the restriction *is* the cycle C_m or C_{2m} , exactly
- ▶ classically there is no walk to ride $\Rightarrow$ reproducing the signal means simulating the circuit

The collapse is not a calculation. It is what the evolution does.

The signal

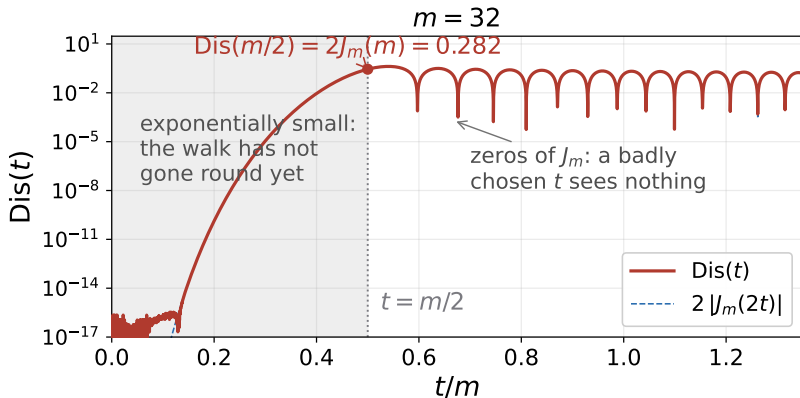
- ▶ a cycle looks the same from every vertex $\Rightarrow$ **flat weights**

$$f_n(t) = \frac{1}{n} \sum_k e^{-i\lambda_k t}, \quad \lambda_k = 2 \cos \frac{2\pi k}{n}$$

- ▶ $\text{Spec}(C_m) \subset \text{Spec}(C_{2m}) \Rightarrow$ the two signals share every term the small cycle has
- ▶ $\Rightarrow$ the difference is one term: the *small* cycle has come back, the big one has not

Uniform weights on an exact cosine grid.

What the difference looks like



$\text{Dis}(t) = |f_{C_m}(t) - f_{C_{2m}}(t)|$ at $m = 32$, on a log scale running down to 10^{-17} . The dashed curve is the *single* Bessel term $2|J_m(2t)|$ that the small cycle adds by coming round early, and the big one does not.

Silent until half way round — then a Bessel function.

Who does what

- ▶ we wrote the circuit, so we know $m \Rightarrow$ **both cycles are known**, with no computation
- ▶ we even know **when to look** — half way round, and there the signal is one Bessel value:

$$t^* = \frac{m}{2}, \quad \text{Dis} = 2 J_m(m) = \Theta(m^{-1/3})$$

- ▶ $\Rightarrow \mathcal{O}(m^{2/3})$ repetitions — polynomial
- ▶ what we do *not* know is **which cycle we are on**

The map is free. Our position on it costs a quantum computer.

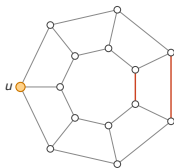
One template

- ▶ a big sparse H , and a starting state its symmetry cannot resolve
- ▶ $\Rightarrow$ the walk stays on a small invariant subspace, by itself
- ▶ $\Rightarrow$ the dynamics is a walk on a small graph A , and the signal is $\langle v | e^{-iAt} | v \rangle$
- ▶ nothing has to be found: no symmetry, no classes, no description of A

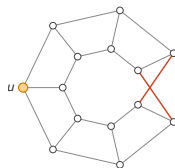
So: which small graphs can we hide this way?

Now we choose what to hide

In BQP the graph was *forced* on us; one reduction, and it makes cycles. Here we choose, so we choose the **hardest** pair we can.



prism Y_7

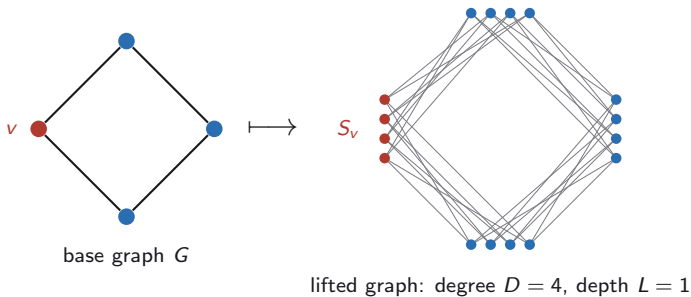


Möbius ladder M_7

- ▶ 3-regular on $n = 2m$ vertices: a ladder closed straight, or twisted
- ▶ two red edges in each: the *entire* difference
- ▶ u : our entry point, kept far from both

Classically: walk it, $O(n)$ steps. So make it hard.

The lift



- ▶ every vertex becomes a **cluster** of D^L copies; every edge of G becomes a **bipartite graph** between the two clusters
- ▶ **regular** — the same number of partners for everyone $\Rightarrow$ uniform over a cluster $\mapsto$ uniform over its neighbour
- ▶ **random**, and every label **shuffled** $\Rightarrow$ degree D is *all* a local observer can read

Y_7 against M_7 this way: $\approx 2 \times 10^{11}$ vertices.

But now we are stuck too

- ▶ after the lift, *being at v* means the uniform state over v 's cluster:

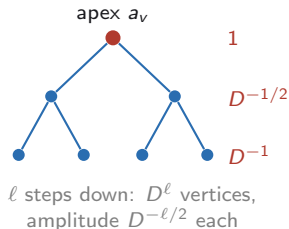
$$|S_v\rangle = D^{-L/2} \sum_{x \in S_v} |x\rangle$$

- ▶ the oracle is a **classical** black box — a label in, its neighbours' labels out; the **same** one serves the classical algorithm
- ▶ $\Rightarrow$ it cannot hand us $|S_v\rangle$; building it ourselves means collecting all D^L shuffled labels — the search we just made hard

To walk on G we need a state we cannot prepare.

The spire is the way in

- ▶ crown each cluster with a **tree** of height L : one **apex** a_v on top
- ▶ the apex is a *single label* — which the oracle *can* hand over
- ▶ $\Rightarrow$ L steps down land exactly on $|S_v\rangle$



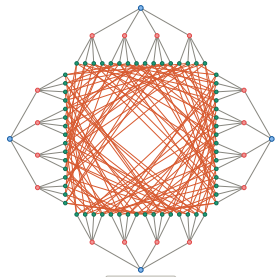
Walking down the spire builds the cluster superposition for free.

Two collapses

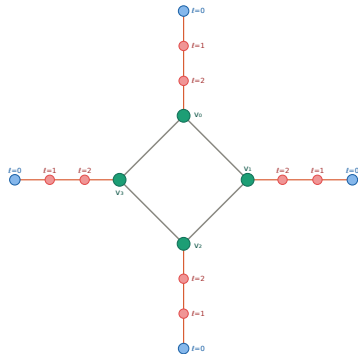
- ▶ the walk only ever sees one whole *level* of a spire at a time
⇒ each **tree** collapses to a **path**: the **tower**, $L = n-1$ edges, all of weight γ
- ▶ it only ever sees one whole *cluster* at a time ⇒ the **lifted graph** collapses to the **base graph** G
- ▶ ⇒ G_{spire} collapses to G_{tower} : the base graph G with a tower hanging from **every** vertex

Two collapses, one walk.

G_{spire} and G_{tower} : the same walk, two sizes



G_{spire} : $\Theta(nD^L)$ vertices, *hidden*



G_{tower} : $n(L+1) = n^2$ vertices, *on paper*

An exponentially large walk, run by an n^2 matrix:

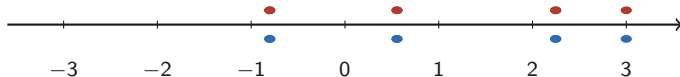
$$H_{\text{spire}}|_{\mathcal{K}} = A_{\text{tower}}$$

So the spectrum we need is G_{tower} 's — we can only hide G through its towered version.

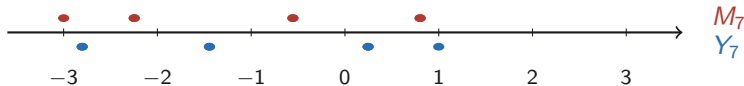
Spectra of the prism and the Möbius ladder

- ▶ the two graphs differ only in their **eigenvalues** — and barely even there

half the spectrum: **identical**



the other half: shifted by **half a spacing**

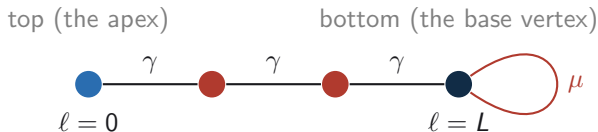


About the smallest difference a walk can still see — and it has to find it.

Same shape as before: a $\mathbb{Z}_2$ **cover** — the twist in the Möbius band, the extra qubit in the circuit.

What the towers do to the spectrum

- ▶ G_{tower} has n^2 vertices $\Rightarrow$ an $n^2 \times n^2$ eigenproblem, $\mathcal{O}(n^6)$ — hopeless at $n \approx 10^4$
- ▶ but the towers act **channel by channel**: one path per base eigenvalue μ_j

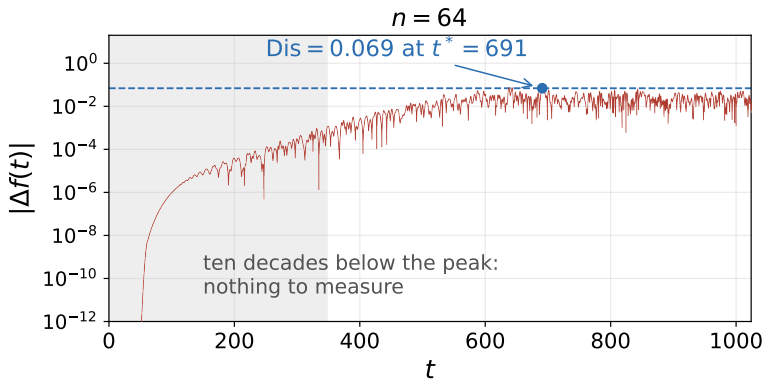


- ▶ n **independent** scalar problems, weights in closed form $\Rightarrow$ the whole signal in $\tilde{\mathcal{O}}(n^2)$

The boundary potential is the hidden graph's eigenvalue.

All classical — and it is what the numerics, and so the conjectures, rest on.

The signal



$|\Delta f(t)|$ for the prism against the Möbius ladder at $n = 64$, on a log scale: exponentially small while the walk is still far from the difference, then a dense band in which t^* is the **highest** point.

Nothing to see, until one particular moment.

Why there is a peak, and what it costs

- ▶ $\Delta f(t) = f_{Y_m} - f_{M_m}$ is about n^2 phases of size about $1/n^2 \Rightarrow$ typically $1/n$ at a random time
- ▶ at t^* enough of them line up: $\text{Dis} = \Theta(\sqrt{\log n} / n)$, and t^* is of order n^2 [NUMERICS, $n \leq 10^4$]

quantum $\sim n^2$ repetitions, $\tilde{O}(n^4)$ gates [CONJ]

classical exponentially many queries [CONJ]

The spires let the walk in, but they are also why t^* has to be found numerically.

What is proved, and what is not

Theorems.

- ▶ the BQP reduction: the walk really is C_m or C_{2m} , and t^* is in closed form
- ▶ the spired construction: one path per base eigenvalue, the prism/Möbius half-step, and the spectral theory throughout

Conjectures.

the two central claims of the paper

- ▶ classical hardness, *relative to the oracle* [Conj. 11.1]
 - ▶ it inherits the **welded-trees** obfuscation, where an exponential classical lower bound *is* proved — but for traversal, not identification
- ▶ the total quantum cost, $\tilde{O}(n^4)$ gates [Conj. 10.1]

The mechanism is rigorous. The *separation* is not.

In some sense, the power of quantum computation is identifying hidden graphs.

Why another black-box problem?

- ▶ exponential separations are rare, and they divide by the *question* asked
- ▶ **hidden subgroup** — *which one is it?* answered with a group to exploit (discrete log, Shor)
- ▶ **welded trees** — *can you get there?* no group, and a proved lower bound
 - ▶ G_{spire} specializes to it at $G = K_2$
- ▶ **here** — *which one is it?* with no group to exploit

All of it is relative to the oracle.

Open:

- ▶ *prove* it for Y_m against M_m — we have extensive numerics, not a proof
- ▶ find this structure arising *on its own*, somewhere in mathematics

Where this comes from

- ▶ **this talk** — *Quantum algorithm for identifying hidden graphs*, arXiv:2605.11228
- ▶ **BQP-complete problems** — Janzing & Wocjan, *Theory of Computing* 2007, *QIC* 2010
- ▶ **spectral graph theory** — with C. Elphick, 2013–2019
- ▶ **welded trees** — Childs et al., *STOC* 2003

The BQP construction is not part of the spired graph paper — this is the context it sits in.